HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed low-level **C** drivers for Qualcomm Snapdragon subsystems, focusing on **driver/firmware development** and validation for new hardware IPs.
- Migrated existing **C** drivers to new hardware platforms, ensuring seamless integration and maintaining functionality across diverse system architectures.
- Implemented **automation scripts** using **Python** to streamline test execution and data collection, improving the efficiency of the validation process.
- Designed and executed comprehensive **test plans** for system hardware validation, covering functional, performance, and integration testing phases.
- Participated in **code refactoring** initiatives to enhance the maintainability and performance of validation software, adhering to best practices.
- Managed the **debug and resolution** of complex issues identified during hardware validation, collaborating with cross-functional teams to identify root causes.
- Containerized validation environments using **Docker**, enabling consistent and reproducible testing setups across different development stages.

SKILLS

Programming Languages – C, C++, Assembly, Python

Processors – ARM, RISC V

Methodologies – Automation, Test/Integration, Validation Methodologies, Test Plan Development

Tools – GIT, Perforce, JIRA, Docker

Domain – Low Level Software Development, Driver/Firmware Development, System Validation, Hardware IP Validation

PROJECTS

Embedded C Driver Development for IoT Devices

- Engineered low-level **C** drivers for a suite of embedded IoT sensors, managing hardware interfaces and data acquisition.
- Developed **automation scripts** in **Python** to facilitate rapid testing and validation of driver functionality, reducing manual effort by approximately 40%.
- Integrated drivers into a simulated RTOS environment, ensuring compatibility and performance under typical operating conditions.

Hardware Validation Framework Automation

- Created a **validation framework** using **C++** to automate hardware testing procedures for new chipsets, supporting both pre-silicon and post-silicon validation.
- Implemented **test execution** logic and reporting mechanisms within the framework, enabling engineers to quickly identify and triage hardware defects.
- Managed source code using **GIT** and tracked defects through **JIRA**, ensuring efficient collaboration and issue resolution within the validation team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)